Bob's Pizza Emporium

Software Design Description

11/11/25

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Version 2.0

The Quintessentials

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Revision History

Date	Version	Description	Author(s)
10/16/2025	1.0	Initial Version	Joshua Swanson, Kelly Byrne, Alison Michailof, Noah Allen, Caelyn Ruiz
11/11/2025	2.0	Second Version: adjusted the mockup GUIs and data flow diagrams	Joshua Swanson, Kelly Byrne, Alison Michailof, Noah Allen, Caelyn Ruiz

1. Introduction

1.1 Purpose

The Purpose of this Software Design Description (SDD) is to describe the system structure, data organization, and design components that implement both the functional and non-functional requirements defined in the Software Requirements Specification (SRS). This document is a guide for developers and testers to understand how the system's architecture and components interact to achieve the goals of Bob's Pizza Emporium. The SDD ensures that the software system design aligns with goals defined in the Software Development Plan (SDP) and maintains consistency throughout development and testing.

1.2 Scope

This design supports a point-of-sale application for pizza businesses, allowing employees to take and manage customer orders, as well as calculate totals with tax. Administrative users can update prices, configure tax rates, modify menu items, and manage user accounts. Whereas "regular" users can only add orders to the ticketing system.

1.3 Definitions, Acronyms, and Abbreviations

Acronym	Meaning
DB	Database
GUI	Graphical User Interface
PIN	Personal Identification Number
SDD	Software Design Description
SDP	Software Development Plan
SRS	Software Requirements Specification
TM	Traceability Matrix

1.4 Document Reference

Document Title	Version	Date	Author(s)

Software Development Plan	1.0	9/18/2025	Joshua Swanson, Noah Allen, Alison Michailof, Kelly Byrne, Caelyn Ruiz
Software Requirements Specification	1.0	10/2/2025	Joshua Swanson, Noah Allen, Alison Michailof, Kelly Byrne, Caelyn Ruiz
Traceability Matrix	1.0	10/9/2025	Joshua Swanson, Noah Allen, Alison Michailof, Kelly Byrne, Caelyn Ruiz

1.5 System Overview

The system will serve as a point-of-sale application for pizza businesses, which will allow employees to take and manage customer orders, in addition to calculating their totals with tax included. It will also allow administrators to update prizes, configure tax rates, modify menu items, and manage user accounts.

2. Data Design

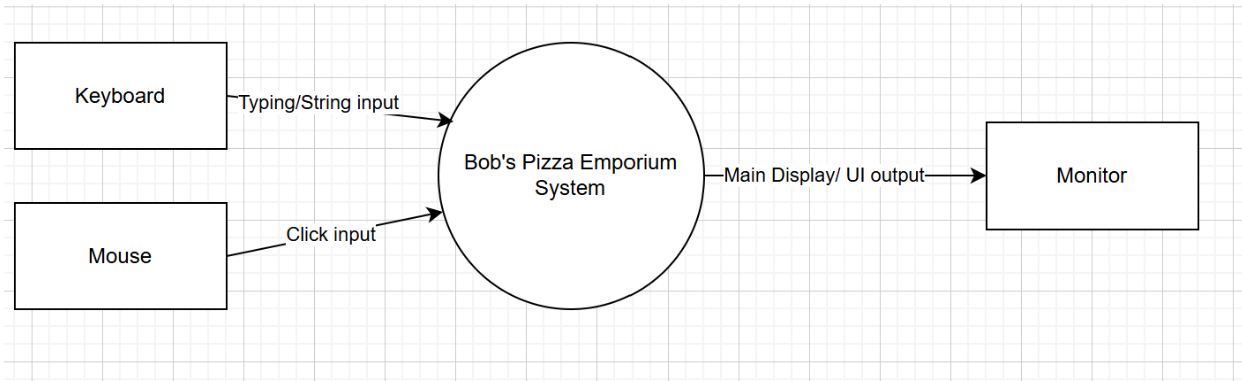
2.1 Internal Data Structures

Bob's Pizza Emporium relies on several core data structures that store and organize information for the users, menu items, and orders that are all stored in a SQLite database.

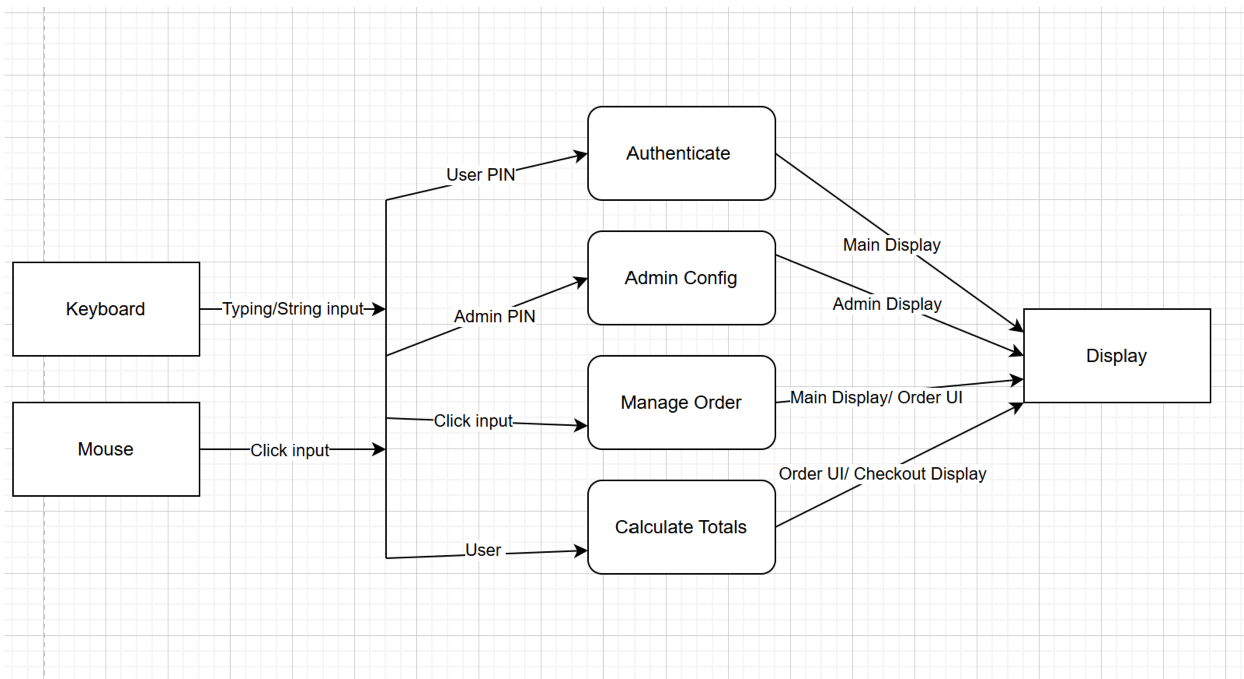
- User Table
  - Stores user credentials and roles
  - Fields: PIN (int, primary key), Username (string), Role (string).
  - Purpose: Provides secure authentication for role-based access.
- Menu Table
  - Contains all pizza and drink items.
  - Fields: ItemID (int, primary key), Price (double).
  - Purpose: Allows for change of menu item prices.
- Toppings Table
  - List customizable topping options.
  - Fields: ToppingID (int, primary key), ToppingName (string).
  - Purpose: Allows customers to customize their pizzas.
- Order Table
  - Tracks the individual customer order and calculates the total.
  - Fields: OrderID (int, primary key), PIN(int, foreign key), OrderDate (datetime), Subtotal (double), Tax (double), Total (double), Status (string).
  - Purpose: Stores order transaction details.
- OrderDetails Table
  - Contains specific items within each order.
  - Fields: DetailID (int, primary key), OrderID (int, foreign key), ItemID (int, foreign key), Quantity (int).
  - Purpose: Links the menu items to their order.

2.2 Data Flow Diagrams

Level 0:



Level 1:



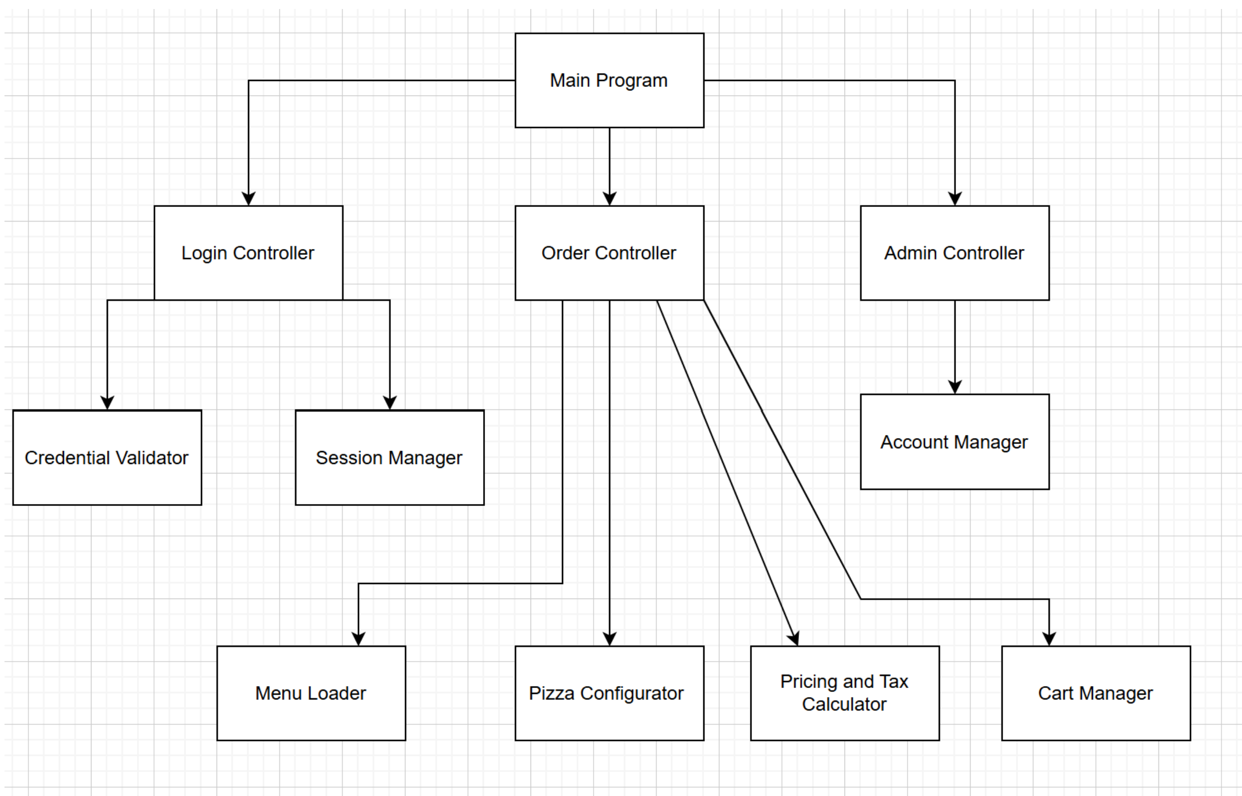
2.3 Data Dictionary

Data Name	Type	Source or Destination	Description
Category	String	Menu Table	Defines the menu group
ItemID	Integer	Menu Table/ OrderDetails Table	Unique ID for each menu item
ItemName	String	Menu Table	Name of item
OrderDate	Datetime	Order Table	Date and time when the order was processed
OrderID	Integer	Order Table/ OrderDetails Table	Unique identifier for each transaction
PIN	Integer(4-digits)	User Table/ Login	An authentication code used for users and a unique identifier for users
Price	Double	Menu Table/ Toppings Table/ Orders Table	Cost for item(s)
Quantity	Integer	OrderDetails Table	Number of units ordered for each item
Role	String	User Table	Determines access level within the system
Status	String	Order Table	Indicates the order state
Subtotal	Double	Order Table	Combined cost of all items before tax

Tax	Double	Order Table	Calculated tax amount
ToppingID	Integer	Toppings Table	Identifier for each topping
ToppingName	String	Toppings Table	Name of topping
Total	Double	Order Table	Final total with tax
Username	String	User Table/ Login	Employee's username

3. Architecture Design

3.1 Program Structure



4. Interface Design

The interface design describes the internal and external program interfaces, as well as the user interface.

4.1 Internal Design

Primary internal interface:

- Login
    - Passes user credentials (PIN) for authentication.
    - Returns a user role identifier to direct users to the correct interface.
- Employee interface
    - Sends selected pizza size, toppings, and drink data.
    - Receives subtotal, tax, and total for order confirmation.
- Admin interface
    - Sends updates to user accounts.
- Ordering Processing
    - Writes new order records.
    - Returns stored order identifiers and receipt details.
- Menu Management
    - Maintains referential consistency between menu and order tables.

4.2 External System Interfaces

External interfaces:

- Operating System interface
    - The application runs on Windows 10 and Windows 11.
    - It relies on system-level libraries for GUI rendering, file handling, and data input/output operations.

These external system interfaces ensure that Bob's Pizza Emporium maintains connections with its host environment and the database for operational efficiency defined in the Traceability Matrix (TM).

4.3 User Interfaces

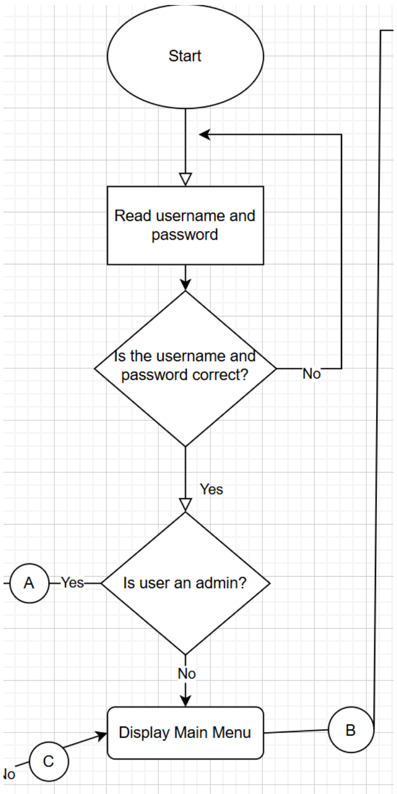

Bob's Pizza Emporium
Enter PIN...
Log In


logout
Home
username
configure order
[ongoing total...]
add a pizza
add a drink
checkout

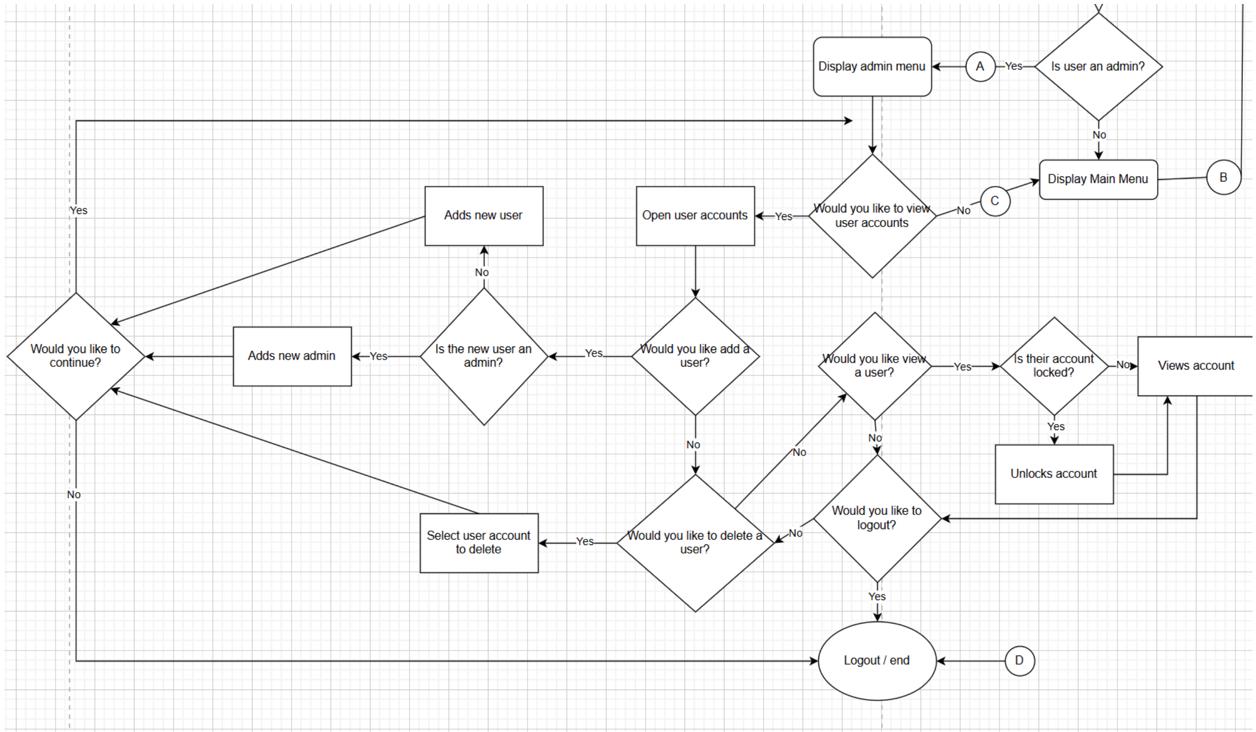

back
Add Pizza
username
Current Pizza
[list as toppings are added...]
Size

5. Procedural Design

Login Menu:

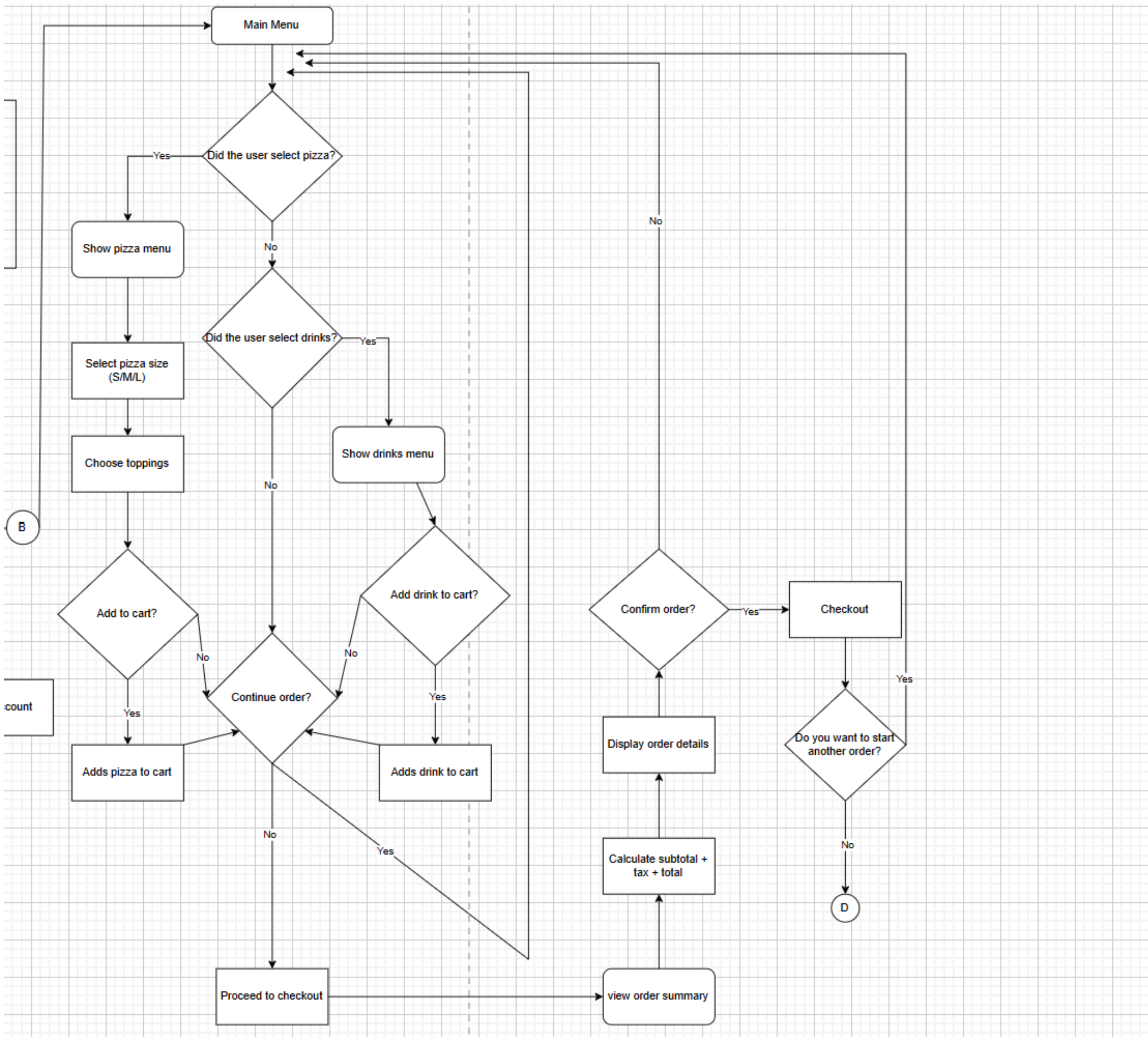


Admin menu:



Display admin menu
A
Yes
Is user an admin?
No
Display Main Menu
B
C
No
Would you like to view user accounts
Yes
Open user accounts
Adds new user
No
Would you like add a user?
Yes
Is the new user an admin?
Yes
Adds new admin
Would you like to continue?
Yes
No
Would you like view a user?
Yes
Is their account locked?
No
Views account
Yes
Unlocks account
No
Would you like to delete a user?
Yes
Select user account to delete
No
Would you like to logout?
No
Yes
Logout / end
D

Ordering menu/checkout menu:



6. Miscellaneous

There are no miscellaneous items at this time.